

MTH209: Worksheet 7

Parallelizations with simulation studies

The library `foreach` and `doparallel` are used for parallelized implementations. Run and study the code below:

```
library(foreach)
library(doParallel)

# Generate large dataset
n <- 1e5      # adjust if required
x <- 1:n

# Parallel mean using foreach

# make and open cluster
num_cores <- parallel::detectCores() - 1
cl <- makeCluster(num_cores)
registerDoParallel(cl)

total_sum <- foreach(i = 1:num_cores, .combine = "+") %dopar% {

  start <- (i - 1) * chunk_size + 1
  end   <- min(i * chunk_size, n)

  sum(x[start:end])/n
}

mean_par <- total_sum

# make sure to stop cluster
stopCluster(cl)
```

1. Suppose $T_1 = \bar{X}$ and $T_2 =$ sample median are two estimators of the central parameter of a t_3 distribution

($\mu = 0$). I say an estimator is better than the other if it has smaller mean squared error:

$$\text{Mean Squared Error} = \mathbb{E}[(T_i - \mu)^2] = \text{Var}(T_i) + [\text{Bias}(T_i)]^2 .$$

- a. For $n = 10$, estimate the MSE for T_1 and T_2 . Which one is better?
 - b. Repeat for larger n . Does your answer change?
 - c. Repeat for $F = N(0, 1)$.
 - d. Write a function: `mean_vs_median()` that takes input `n.vec` and returns estimates of MSE of mean and MSE of median for one replication for all the grid of values in `n.vec`.
 - e. Find average MSE for 1000 replications for `n.vec <- c(10, 50, 100, 200, 500, 1000, 2000, 5000)`.
2. Repeat Question 1 using parallelizations with `foreach` and `doParallel` functions. Which code is faster?
3. Verify that `max` and `min` are embarassingly parallel, but `median` is not.
4. Consider a regression model where for $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, and $\beta \in \mathbb{R}^p$, we have

$$y \sim N(X\beta, \sigma^2),$$

for $\sigma^2 > 0$.

- a. Generate data from this model using some true values of β and σ^2 and X with $n = 1e3$ and $p = 1e2$.
 - b. Obtain the regression estimator: $\hat{\beta} = (X^\top X)^{-1} X^\top y$.
 - c. Estimate the average MSE of $\hat{\beta}$: $\mathbb{E}[\|\hat{\beta} - \beta\|^2]$ using 1000 replications.
5. Repeat 4. using `foreach`.
6. Verify that the regression estimator is not embarassingly parallel.